The Relational Universe: Interaction and the Transition of the Possible to the Actual

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Abstract

We argue that quantum randomness is a logical necessity arising from the nature of intrinsically relational properties—properties lacking definite states outside specific contexts of causal interaction. This motivates Relational Actualism (RAQM), an ontological framework classifying physical properties as invariant, classical, or intrinsically relational. RAQM grounds wave function actualization in physical interaction rather than observer-relative facts. The actualization criterion is physically grounded and mathematically precise: an interaction constitutes an actualization event when it produces an irreversible increase in the quantum relative entropy of the system with respect to the vacuum reference state, $\Delta S(\rho\|\sigma_0) > 0$, generating a robust, redundant causal record. The on-shell condition $p^\mu p_\mu = m^2c^2$ is the implementation of this entropic criterion in weakly coupled perturbative QFT; it is not the fundamental definition and does not constitute a claim of sharp spacetime localization (see Section 3.2.1 for the explicit response to Reeh-Schlieder and LSZ concerns). This criterion is basis-independent, non-perturbative, and requires no new fundamental constants. To reconcile a globally consistent wave function with special relativity, the Tomonaga-Schwinger equation establishes a foliation-independent global update mechanism. We derive a consistency condition demonstrating that RAQM's actualization map preserves identical observable statistics at spacelike-separated points regardless of hypersurface choice—a derived result, not a postulate—strictly forbidding superluminal signaling via algebraic microcausality. The no-signaling result applies rigorously to the non-selective (outcome-averaged) actualization map; conditioned branches are treated explicitly. Three specific, discriminating empirical predictions are stated: a finite actualization propagation velocity bounded by the Lieb-Robinson speed, an environment-dependent transition duration scaling as $\hbar/\Delta E_{\text{env}}$, and a sharp reversibility threshold at the actualization criterion. RAQM is distinguished from Relational Quantum Mechanics (which relativizes facts), dynamical collapse models (which ground collapse in stochastic gravitational thresholds rather than causal interaction), decoherence-only models (which leave the outcome problem open), and Everettian Many-Worlds (which denies definite events). Classical behavior emerges dynamically when continuous actualization events eliminate all unconstrained degrees of freedom. The companion papers RAGC and RACI extend these results to gravity, cosmology, and biological complexity; RAHC [Sandeman, 2026d] extends the framework further to the emergence of agentic intelligence, formalizing the recursive coarse-graining hierarchy that generates successive levels of causal closure from the microscopic DAG; key theorems in those papers have been formally verified in Lean 4/Mathlib. A second Lean 4 file (`RA_AQFT_Proofs_v10.lean`) formalises the finite-dimensional AQFT proof program and proves the frame-independence and Rindler stationarity theorems; see the note on formal verification in the companion paper RAGC [Sandeman, 2026b].

Keywords: Relational Quantum Mechanics · Wave Function Collapse · Microcausality · Decoherence · Determinism · Phase Space · Objective Collapse · Entanglement Entropy · Lean 4 Verification

1 Introduction

If the universe is, at least under certain circumstances, random, then it is impossible, even in principle, to build a deterministic model of it. In the best case, it can yield possible outcomes and their probabilities of occurring. What actually happens must be added to the model. As far as we know, there are indeed circumstances in which the universe is truly random. Furthermore,

randomness in the universe is not some vexing and curious add-on; it is, in fact, logically necessary—a consequence, as we shall see, of the relational nature of the universe.

The paper proceeds as follows. Section 2 situates the view philosophically, distinguishing it from other interpretations including Relational Quantum Mechanics, dynamical collapse models, and decoherence-based approaches. Section 3 develops the mathematical formalism in terms of density matrices, decoherence, and the reduced density matrix. Section 4 walks through the key experimental evidence. Section 5 discusses the broader implications for determinism, causality, the reality of the wave function, and empirical falsifiability. Section 6 concludes.

2 Rethinking Quantum Theory: Interpretive Landscape

Erwin Schrödinger once said of Quantum Theory, "I don't like it, and I'm sorry I ever had anything to do with it." What Schrödinger objected to—which inspired his famous thought experiment involving a hapless cat—was randomness. Einstein shared his concerns. Quantum Theory has the dubious distinction of being one of the most powerful and accurate models of the world ever developed, while also being the source of considerable controversy, bewilderment, and confusion. The root of the problem is that, under certain circumstances, Nature appears to be truly random, which necessitates a significant shift in thinking about how it can be modeled and how it evolves.

2.1 A Taxonomy of Properties

Phenomena in nature have a set of possible states, only one of which can be actualized (manifest, measurable) at any given point in spacetime. Properties can be categorized as follows:

Invariant properties are those that can only ever have one possible state—e.g., the charge of an electron. Their probability of measurement in the one state is unity.

Classical properties are either fixed or vary deterministically as the result of deterministic interactions. They are actualized as a state of being. The trajectory of a planet orbiting in a gravitational field is a paradigm case.

Intrinsically relational properties cannot assume definite values outside a specific context of interaction. Examples include the location or spin of newly created particles whose spins are correlated. Crucially, after undergoing an appropriate interaction, intrinsically relational properties can become classical—they become actualized and behave the same as invariant properties, so long as the context of interactions eliminates all other possibilities.

The Special and General Theory of Relativity introduced the world to the concept of relational properties, demonstrating that there is no absolute coordinate system for spacetime. Quantum mechanics goes further, showing that intrinsically relational properties, prior to an interaction that restricts the state space, are multivalued. In Quantum Theory this is modeled as a linear combination of amplitudes which, via the Born Rule, correspond with very high accuracy to the probability that a certain state will be actualized given a particular type of interaction.

2.2 Positioning: Relational Actualism vs. Relational Quantum Mechanics

Carlo Rovelli's Relational Quantum Mechanics (RQM) shares with the present framework the conviction that quantum states are not absolute but are defined relative to an observer or reference system [Rovelli, 1996]. In RQM, there are no observer-independent facts about the values of physical quantities; every fact is a fact relative to some system.

The present view, Relational Actualism, agrees that quantum properties are relational and context-dependent but diverges from RQM in three important respects. First, where RQM relativizes facts to observers, Relational Actualism grounds actualization specifically in causal interaction: a property becomes actual when an interaction permanently entangles the system with its asymptotic environment and generates an irreversible causal record. The notion of 'observer' is not doing fundamental work here—every physical interaction that meets this threshold constitutes an actualization event.

Second, RQM is comfortable with different observers assigning incompatible wave functions to the same system. The present view holds, by contrast, that there is a single globally consistent wave function for the universe—following Hartle and Hawking [Hartle and Hawking, 1983]—and that apparent perspectival differences reflect the local availability of information rather than genuine ontological relativity.

Third, Relational Actualism makes a stronger ontological claim about Hilbert space: the amplitudes and their interference structure are physically real, not merely predictive bookkeeping. This claim is supported by the Berry phase, interference washout experiments, and the results of the quantum eraser discussed in Section 4.

Table 1: Interpretive landscape: Comparing RAQM with major alternatives.

Framework	**Global State**	**Facts**	**Actualization**	**Randomness**
Everett [Everett, 1957]	Yes	Relative to branch	None (unitary branching)	Epistemic
Zurek [Zurek, 2003]	Yes	Left ambiguous	Apparent (einselection)	Epistemic
GRW/CSL [Bassi et al., 2013]	Yes	Objective	Stochastic gravitational/ noise threshold	Ontic
Rovelli [Rovelli, 1996]	No	Relative to observer	Perspectival interaction	Ontic
RAQM (This)	Yes	Objective (post-event)	Causal pruning (entropy criterion)	Ontic

2.3 Decoherence and Zurek's Einselection: Engagement and Distinction

Decoherence theory, developed most systematically by Zurek et al., provides the dominant technical framework for explaining the quantum-to-classical transition [Zurek, 1981, 2003, Joos and Zeh, 1985]. The central idea is that quantum systems interact with environmental degrees of freedom, causing the reduced density matrix to rapidly lose its off-diagonal coherences in a preferred basis—the pointer basis. This process—einselection—explains why macroscopic objects appear classical.

Decoherence theory is mathematically rigorous and empirically well-confirmed. However, it faces a residual interpretive problem: decoherence explains why interference terms become practically unobservable, but it does not explain why a single outcome occurs. The full density matrix of system-plus-environment remains in a superposition; only the reduced density matrix loses coherence. Relational Actualism offers a resolution: the loss of coherence tracked by decoherence reflects a genuine causal process in which environmental interactions eliminate possibilities from Hilbert space. Decoherence theory correctly describes the mathematical structure of this process; the present view adds the ontological interpretation that possibility-elimination is real and that the resulting actualized state is genuinely definite, not merely effectively so. The wave function of the universe is not branching, but pruning.

2.4 Positioning Against Dynamical Collapse Models

Dynamical collapse models—most prominently the Ghirardi-Rimini-Weber (GRW) theory, Continuous Spontaneous Localization (CSL), and the gravity-induced collapse models of Diósi and Penrose—share RAQM's commitment to objective, physical collapse [Bassi et al., 2013]. In these frameworks, collapse is a real physical event, not merely an epistemic update. RAQM is in broad agreement with this ontological commitment and regards these models as pointing in the right direction.

However, RAQM diverges from dynamical collapse models in two important respects. First, the collapse trigger: GRW and CSL invoke a stochastic noise field with a new fundamental constant (the collapse rate), while Diósi-Penrose invoke a gravitational threshold. RAQM instead grounds actualization in the causal structure of physical interaction itself—specifically, in the irreversible increase

of entanglement entropy with the asymptotic environment produced by an on-shell interaction. This criterion requires no new fundamental constant and is basis-independent by construction (defined at the level of the density matrix, not a perturbative expansion). A full claim of gauge-invariance requires additional care in theories with constrained Hilbert spaces and edge modes, as discussed in Section 3.2.1; this is an open refinement.

Second, the domain of application: dynamical collapse models are designed to solve the measurement problem by modifying the Schrödinger equation at a specific scale. RAQM does not modify the dynamics; it provides an ontological interpretation of what the existing formalism tracks. The entropy criterion is not an additional term in the Hamiltonian but a physical condition distinguishing actualizing from non-actualizing interactions within the existing QFT framework. For a comprehensive review of dynamical collapse models and their experimental status, see Bassi et al. [Bassi et al., 2013].

3 Mathematical Formalism

3.1 Pure States, Superposition, and the Born Rule

A quantum system is described by a state vector $|\psi\rangle$ in a Hilbert space $\mathscr{H}$. For a two-state system:

$$|\psi\rangle = \alpha|x_1\rangle + \beta|x_2\rangle, \qquad \alpha,\beta \in \mathbb{C}, \quad |\alpha|^2 + |\beta|^2 = 1. \tag{1}$$

The coefficients α and β are probability amplitudes. By the Born Rule, the probability of finding the system in state $|x_i\rangle$ upon measurement is $|\langle x_i|\psi\rangle|^2$. Prior to a context-defining interaction, both terms in the superposition are physically real in the sense that they contribute to interference phenomena. The double-slit experiment exemplifies this directly: the detection probability $|\psi(y)|^2$ contains an interference term $2\,\mathrm{Re}(\langle y|A\rangle\langle B|y\rangle)$ responsible for the fringe pattern, which vanishes exactly when which-path information is available in the environment.

3.2 Density Matrices and the Reduced Density Matrix

The density matrix formalism is essential for describing open quantum systems. For a pure state $|\psi\rangle$, the density matrix is $\rho = |\psi\rangle\langle\psi|$. The off-diagonal terms $\alpha\beta^*$, $\alpha^*\beta$ are the coherences—the mathematical signature of genuine quantum superposition. Consider the total system: particle S plus environment E. If the environment records which-path information, the composite state evolves as:

$$|\Psi_{\text{total}}\rangle = \alpha|x_1\rangle|\varepsilon_1\rangle + \beta|x_2\rangle|\varepsilon_2\rangle, \tag{2}$$

where $|\varepsilon_1\rangle$, $|\varepsilon_2\rangle$ are environmental states correlated with paths x_1, x_2. The reduced density matrix of S is obtained by tracing over E:

$$\rho_S = \mathrm{Tr}_E\big(|\Psi_{\text{total}}\rangle\langle\Psi_{\text{total}}|\big) = |\alpha|^2|x_1\rangle\langle x_1| + |\beta|^2|x_2\rangle\langle x_2| + \alpha\beta^*\langle\varepsilon_2|\varepsilon_1\rangle|x_1\rangle\langle x_2| + \text{c.c.} \tag{3}$$

The off-diagonal terms are suppressed by $\langle\varepsilon_2|\varepsilon_1\rangle$—the overlap between environmental states. When $\langle\varepsilon_2|\varepsilon_1\rangle = 0$, the reduced density matrix becomes diagonal—indistinguishable from a classical probability distribution. Relational Actualism interprets this as follows: the environmental interaction has eliminated the non-local path possibilities from the particle's portion of Hilbert space. The coherences $\langle\varepsilon_2|\varepsilon_1\rangle \to 0$ is the mathematical signature of a genuine actualization event.

3.2.1 The Actualization Criterion

For Relational Actualism, the actualization criterion is: *an interaction constitutes an actualization event when it produces an irreversible increase in the entanglement entropy of the system with its asymptotic environment*—generating a robust, redundant causal record recoverable in principle by any future observer in the causal future of the interaction. This criterion is *sufficient* for actualization across all regimes treated here. It is also *necessary* in specific well-controlled models—Markovian

open systems and perturbative QFT with on-shell emission into the far field—where the connection between environmental orthogonality and irreversible record formation can be made explicit. Whether necessity holds in full generality is an open problem; a complete proof would require a factorization-independent, algebraic-QFT formulation of irreversibility—for instance, via relative entropy monotones on local von Neumann algebras.

This criterion is basis-independent: entanglement entropy $S(A) = -\mathrm{Tr}(\rho_A \log \rho_A)$ does not depend on the choice of computational basis for the system's Hilbert space, and it is defined at the level of the density matrix rather than the Feynman diagram expansion, making it non-perturbative. A stronger claim of full gauge-invariance requires additional care. In gauge theories, Hilbert-space factorization and entanglement depend on the choice of operator algebra and the treatment of edge modes at the boundary between system and environment—so 'basis-independent' does not automatically entail 'gauge-invariant' in constrained systems. The criterion is gauge-invariant in the sense that it does not depend on a choice of perturbative gauge fixing or bookkeeping; extending it rigorously to theories with non-trivial edge mode structure (electrodynamics, non-Abelian gauge theories) is an open refinement, likely requiring the algebraic-QFT formulation noted above.

In weakly coupled QFT, on-shell emission into the asymptotic environment is a *sufficient* condition for actualization: a real, on-shell excitation satisfying $p^\mu p_\mu = m^2c^2$ propagates into the far field, producing environmental states that are orthogonal (or near-orthogonal) and thereby generating a lasting, redundant entanglement record. Purely virtual (off-shell) exchanges—which mediate van der Waals forces, the Casimir effect, and similar phenomena—typically leave the environment in its ground state and produce only reversible dynamic shifts (such as the Lamb shift) without generating a persistent causal record. However, the on-shell condition is *not* asserted to be necessary: near-field couplings and many-body scattering processes can in principle produce robust environmental distinguishability without a single propagating on-shell quantum, and the physically relevant question in all regimes is whether environmental degrees of freedom end in near-orthogonal states that persist. The on-shell condition is the clean limiting expression of the actualization criterion in radiative perturbative regimes, not its general definition.

This reformulation also handles cases where the on-shell language becomes ambiguous. In strongly correlated condensed matter systems, a quasiparticle excitation—a phonon, magnon, or polariton—constitutes an actualization event when it produces an irreversible entanglement increase in the condensed matter environment, regardless of whether it maps cleanly onto a fundamental gauge boson. Similarly, contact interactions and near-field couplings actualize when they satisfy the entropy criterion, even in the absence of propagating radiation. The deeper requirement in all regimes is irreversible environmental entanglement.

The Kinematic Snap: the microscopic mechanism of irreversibility. The precise physical mechanism that converts reversible potentia into irreversible actuality can be stated in standard QFT terms without appeal to thermodynamic avalanches or statistical thresholds. We call this the *Kinematic Snap*.

When two fundamental particles interact, their wavefunctions (potentia) overlap. At short range, this overlap is mediated by virtual, off-shell bosons satisfying $p^\mu p_\mu \neq m^2c^2$. Virtual bosons are off-shell precisely because they represent the universe's provisional, reversible exploration of possible states: they cannot propagate to the asymptotic environment, so they cannot balance the Local Ledger, and they write no vertex to the DAG. The interaction remains reversible for as long as only virtual exchange occurs.

Actualization occurs at the kinematic threshold: the precise moment at which the interaction possesses sufficient energy-momentum to elevate a mediating boson to the mass shell. At this threshold, the boson transitions to an on-shell real excitation satisfying $p^\mu p_\mu = m^2c^2$. An on-shell boson can propagate to infinity. The Local Ledger Condition then demands permanent recording: the energy-momentum transfer is inscribed as an irreversible vertex on the causal DAG. The continuous,

reversible hum of virtual off-shell potentia snaps into a discrete, permanent causal fact.

Relationship to QFT localization theorems. A natural technical concern is whether identifying actualization with the on-shell threshold contradicts the Reeh-Schlieder theorem or the failure of Newton-Wigner localization at sub-Compton scales. It does not, for two reasons.

First, the RA actualization criterion is *not* a sharp spacetime localization claim. The primary criterion is an irreversible increase in quantum relative entropy $\Delta S(\rho\|\sigma_0) > 0$ with respect to the vacuum reference state (Section 3.2.1). The on-shell condition $p^\mu p_\mu = m^2c^2$ is the implementation of this entropic criterion in the perturbative regime — it is the signature of an excitation that can propagate to the asymptotic environment and generate a redundant causal record. It is explicitly *not* a claim that the interaction is localized at a mathematical point x_0^μ.

Second, the LSZ reduction formula places asymptotic external states on-shell at $t \to \pm\infty$. RA uses the on-shell condition as an effective diagnostic of whether a given exchange has crossed the threshold for environmental entanglement — not as a claim that LSZ applies locally at each vertex in a finite interaction region. The physically relevant object is the S-matrix pole, which is frame-independent and well-defined. What RA adds is the identification of pole-crossing with the writing of a causal DAG vertex: a metaphysical claim about what the S-matrix pole *means*, not a departure from the mathematics of interacting QFT.

In strongly coupled regimes where the quasi-particle picture breaks down, the entropic criterion takes over directly: actualization is the irreversible increase in $\Delta S(\rho\|\sigma_0)$, and the on-shell language becomes an approximation. This is an open refinement (Hard Wall ii in RAGC RAGC Hard Wall ii), not a contradiction.

This mechanism has four immediate consequences:

1. **Irreversibility is kinematic, not statistical.** The threshold is the dispersion relation $p^\mu p_\mu = m^2c^2$—a binary condition grounded in standard QFT, not a probabilistic or thermodynamic criterion. The snap is sharp and objective.

2. **No observer is required.** An electron and a proton do not require a scientist to actualize their binding. Once their interaction produces a real on-shell photon carrying away the binding energy (~ 13.6 eV for hydrogen), the vertex is written regardless of whether anyone is watching. The mechanism is entirely relational and causal.

3. **Finite decoherence time is explained.** Decoherence takes a finite duration not because of a fuzzy statistical process, but because it takes physical time for emitted on-shell bosons to propagate outward and entangle with the surrounding environmental degrees of freedom. The decoherence timescale $\tau_D \sim \hbar/\Delta E_{\text{env}}$ (Prediction P2) is precisely the light-travel time for on-shell radiation to reach the environmental boundary, scaled by the interaction strength.

4. **Virtual exchanges are not actualized.** Van der Waals forces, the Casimir effect, the Lamb shift—all mediated by off-shell virtual exchange—produce no DAG vertices. They are genuine physical effects, but they are effects of the potentia, not actualities. This is consistent with the RA vacuum prescription: the Casimir energy difference (arising from the actualized boundary conditions imposed by the plates) gravitates; the absolute virtual zero-point energy does not.

The Unruh vacuum and the stationarity criterion. The Kinematic Snap also resolves the Unruh frame-dependence question precisely. A uniformly accelerated (Rindler) observer detects a stationary thermal bath at temperature $T_U = \hbar a/(2\pi c k_B)$. This bath is in thermal equilibrium: its relative entropy with respect to the Rindler vacuum is constant in time ($\Delta S_{\text{rel}} = 0$). No irreversible increase in relative entropy occurs; no on-shell bosons are being emitted into the asymptotic environment at a net rate that exceeds absorption. The Rindler thermal bath represents the universe's

stationary potentia as seen by the accelerated observer—it is not generating new actualization events. Both the inertial and the accelerated observer therefore agree: in the empty Minkowski vacuum, no actualization is occurring. The apparent disagreement on particle content is a modal decomposition artefact, not a disagreement about physical causal facts.

This becomes dramatic at the extreme limit. As acceleration—or equivalently, by the Einstein Equivalence Principle, gravitational mass-density—approaches the bandwidth limit $R_{\max}$, the local actualization density of the DAG saturates. The graph cannot accept further causal links. At this limit, the kinematic threshold is reached at every interaction simultaneously: the graph geometrically shears, producing a causal severance. This is the RA account of what standard GR calls a singularity. There is no infinity; there is a graph cut, governed by the same Ledger Partition Theorem verified in Lean 4/Mathlib, forcing the nucleation of a new, causally isolated causal ledger. Recent experiments measuring the finite duration of quantum transitions [Minev et al., 2019] are uniquely positioned to probe the kinematic snap mechanism at the laboratory scale, as discussed in Section 5.5.

Frame-independence of the actualization criterion. A subtle but important point concerns the Lorentz covariance of the criterion. If actualization were defined in terms of particle number—"an on-shell particle is emitted"—it would be frame-dependent: an accelerated (Rindler) observer detects a thermal bath of particles in the Minkowski vacuum, while an inertial observer detects none. This would make actualization observer-dependent, which is precisely what RA rejects.

The entropy criterion resolves this. Relative entropy $S(\rho\|\sigma)$ between states is a diffeomorphism-invariant scalar. A strictly irreversible increase in relative entropy with respect to the vacuum state is an objective, coordinate-independent fact. An accelerated and an inertial observer may disagree on the particle content of an interaction, but they must agree on whether an irreversible thermodynamic event occurred, because irreversibility is encoded in the monotone properties of relative entropy—a frame-independent quantity. The on-shell particle language is the clean limiting expression of this criterion in weakly coupled, perturbative regimes; the fundamental criterion is the relative entropy increase, which is covariant in all regimes. The formal proof that $\mathscr{P}_{\text{act}}$, reformulated in terms of local algebras and modular objects rather than Fock-space particles, produces a frame-independent set of actualization events is pursued in the companion paper RAGC [Sandeman, 2026b].

3.2.2 Lemma: Conservation of Probability under Actualization

When an actualizing interaction occurs, the continuous pruning of the state space must strictly conserve total probability. We establish this in two steps, distinguishing the unconditional update from the post-selection formula.

Step 1: The unconditional update. Let $\{M_k\}$ be the set of Kraus operators representing the environmental scattering channels, satisfying the completeness relation $\sum_k M_k^\dagger M_k = I$. The unconditional (pre-selection) state after the interaction is:

$$\rho_{\text{after}} = \sum_k M_k \, \rho_{\text{before}} \, M_k^\dagger. \tag{4}$$

Probability conservation follows directly from the completeness relation:

$$\mathrm{Tr}(\rho_{\text{after}}) = \mathrm{Tr}\Big(\sum_k M_k \, \rho_{\text{before}} \, M_k^\dagger\Big) = \mathrm{Tr}\Big(\rho_{\text{before}} \sum_k M_k^\dagger M_k\Big) = \mathrm{Tr}(\rho_{\text{before}}) = 1. \tag{5}$$

This establishes that the total probability is preserved *across all possible outcomes*. No probability leaks from the system during the transition from the possible to the actual.

Step 2: The post-selection formula. Once a specific outcome k is realized—i.e., once actualization selects a particular environmental channel—the updated state is renormalized to reflect the realized branch:

$$\rho_{\text{after}|k} = \frac{M_k \, \rho_{\text{before}} \, M_k^\dagger}{\mathrm{Tr}(M_k \, \rho_{\text{before}} \, M_k^\dagger)}. \tag{6}$$

The denominator $\mathrm{Tr}(M_k \rho_{\text{before}} M_k^\dagger) = p_k$ is exactly the probability of outcome k, as given by the Born Rule. The post-selection formula geometrically rescales the remaining phase space to the realized branch, guaranteeing $\mathrm{Tr}(\rho_{\text{after}|k}) = 1$. The laws of probability remain strictly intact. The unconditional update (Step 1) establishes conservation across all outcomes; the post-selection formula (Step 2) gives the state conditional on a specific actualization having occurred.

3.3 Decoherence Timescales

The decoherence timescale τ_D quantifies how rapidly a superposition loses coherence through environmental interaction. For a particle of mass m in thermal contact with an environment at temperature T:

$$\tau_D \approx \tau_R \left(\frac{\lambda_{dB}}{\Delta x} \right)^2, \tag{7}$$

where τ_R is the relaxation time, $\lambda_{dB} = \hbar/\sqrt{2mkT}$ is the thermal de Broglie wavelength, and Δx is the spatial separation of the superposed states. For macroscopic systems, $\Delta x \gg \lambda_{dB}$, and $\tau_D \ll \tau_R$—decoherence occurs many orders of magnitude faster than dissipation. For a buckyball (C_{60}, mass $\approx 1.2 \times 10^{-24}$ kg) at room temperature with path separation $\Delta x \approx 1\,\mu$m, τ_D is of order 10^{-19} s.

3.4 Entanglement Entropy as a Measure of Relationality

The entanglement entropy provides a quantitative measure of how 'relational' a bipartite system is. For a bipartite state ρ_{AB}, the entanglement entropy of subsystem A is:

$$S(A) = -\mathrm{Tr}(\rho_A \log \rho_A), \tag{8}$$

where $\rho_A = \mathrm{Tr}_B(\rho_{AB})$. $S(A) = 0$ if and only if the global state is a product state (A's properties are intrinsic). $S(A)$ is maximized for a maximally entangled state, in which A's properties are entirely relational—undefined without reference to B. This gives precise mathematical content to the taxonomy of Section 2: invariant properties correspond to $S = 0$ for the relevant degree of freedom; intrinsically relational properties correspond to high-$S(A)$ states. The decoherence process continuously drives $S(A)$ upward through entanglement with the environment while reducing local superposition—this is the Hilbert space picture of classicalization. The formal account of how recursive coarse-graining over the microscopic DAG generates successive levels of classical description — from chemistry to biology to agentic intelligence — is developed in the companion paper RAHC [Sandeman, 2026d].

We can map this process directly onto the geometry of a dynamical phase space. In Relational Actualism, entanglement entropy functions as the precise quantitative measure of accumulated causal constraints. Each actualization event physically constrains the system, binding its future evolution to the state of the environment. As $S(A)$ increases, the unconstrained volume of the local quantum state shrinks, funneling the system's trajectory until it is strictly bounded by its relational network.

3.5 The Berry Phase as Evidence for the Reality of Hilbert Space

The Berry phase provides particularly compelling evidence that the geometry of Hilbert space has physical consequences. When a quantum system's Hamiltonian is slowly varied along a closed path C in parameter space, the state acquires a geometric phase:

$$\gamma_n(C) = i \oint_C \langle n(\mathbf{R}) | \nabla_{\mathbf{R}} | n(\mathbf{R}) \rangle \cdot d\mathbf{R}. \tag{9}$$

This geometric phase depends only on the geometry of the path in parameter space—specifically, on the solid angle subtended by C in the projective Hilbert space—not on the rate of traversal or the energy of the state. It has been experimentally confirmed in systems ranging from photon polarization to neutron interferometry to topological insulators [Cohen et al., 2019]. The Berry

curvature $\Omega_n = \nabla_{\mathbf{R}} \times \mathbf{A}_n$ (where $\mathbf{A}_n = i\langle n|\nabla_{\mathbf{R}}|n\rangle$ is the Berry connection) is physically real and measurable, strongly supporting the central ontological claim of Relational Actualism: Hilbert space amplitudes represent real structures whose geometry constrains and shapes physical processes.

4 Experimental Analysis

4.1 The Double-Slit Experiment

The double-slit experiment and its variants provide the central experimental scaffold for the present argument. In the scenario with one open slit, the pattern on the detection screen is consistent with point-like entities. With two slits, a wave interference pattern appears. In the density matrix framework, this is precisely the situation of non-zero off-diagonal coherences: $\langle \varepsilon_2|\varepsilon_1\rangle \approx 1$, because no which-path information has been recorded in the environment. When a detector is placed at one slit, the environment acquires which-path information, $\langle \varepsilon_2|\varepsilon_1\rangle \to 0$, and the interference pattern vanishes. Whether the detector 'fires' or not is irrelevant—what matters is whether the environmental entanglement entropy has increased irreversibly.

4.2 The Buckyball Experiment and the Washout Effect

The buckyball (C_{60}) interference experiment of Hackermüller et al. [Hackermüller et al., 2004] is particularly illuminating because it demonstrates that the transition from quantum to classical behavior is continuous, not discontinuous, and is governed by information content. At low temperatures, buckyballs exhibit clear wave interference through a nanograting. As temperature increases, the buckyballs emit thermal radiation—photons whose wavelength encodes which-path information. At a critical temperature the interference pattern washes out entirely. A variation by Hornberger et al. [Hornberger et al., 2003] achieved the same washout by degrading the vacuum.

The key physical quantity is the overlap $\langle \varepsilon_2|\varepsilon_1\rangle$ between environmental states correlated with the possible paths. The washout is continuous and quantitative, matching the gradual reduction of off-diagonal elements in the reduced density matrix. The energy threshold $E_{\text{wash}} \approx hc/d$ (for slit separation $d \approx 100\,\text{nm}$ gives $E_{\text{wash}} \approx 12.4\,\text{eV}$) is consistent with experimental results.

4.3 Delayed-Choice and Quantum Eraser Experiments

The delayed-choice experiment (Wheeler) demonstrates that the presence or absence of an interference pattern is determined by the total informational state of the system at the moment of detection—the relevant quantity is $\langle \varepsilon_2|\varepsilon_1\rangle$ evaluated over the full system history. This explains why there is no retrocausal signaling.

The quantum eraser experiment [Scully and Drühl, 1982] completes the picture: by destroying which-path information (via a second polarizer), coherence is restored—the off-diagonal terms re-emerge in the reduced density matrix of the signal photons. This demonstrates bidirectionality: actualization can be 'undone' if the environmental record is erased *before* a macroscopic thermodynamic record is irreversibly established. This is critical: the quantum eraser does not reverse an actualization; it reveals that no actualization had yet occurred, because the thermodynamic horizon had not been crossed. This is strong evidence that irreversible environmental entanglement—not the physical presence of a detector—is the operative quantity.

4.4 Quantum Maze: Entanglement and Nonlocal Constraints

The quantum maze experiment uses parametric down-conversion to produce entangled photon pairs. Because the pairs are entangled, they constitute a single quantum system; a measurement on the idler photon instantaneously constrains the state of the signal photon regardless of spatial separation. The present framework interprets this as follows: the entangled pair has a joint wave function that is non-separable. When the idler is detected, the actualization criterion is satisfied and the signal photon's reduced density matrix loses its off-diagonal terms. This nonlocality is not a violation of relativity because no information is transmitted faster than light; both photons belong to a single

nonlocal quantum system whose wave function is defined in the configuration space of both particles jointly.

5 Discussion

5.1 The Many-Worlds Rebuttal

The Everettian Many-Worlds Interpretation (MWI) [Everett, 1957, Saunders et al., 2010] holds that the universal wave function never collapses and evolves unitarily always. MWI has notable virtues: it takes the formalism of quantum mechanics at face value and requires no additional collapse postulate. However, it faces several serious difficulties.

First, the preferred basis problem. While einselection provides a basis-selection mechanism via decoherence, branching must occur in some basis. The pointer basis selected by decoherence depends on the structure of the environment, which must be specified. The branches of MWI are not fundamental features of the wave function but artifacts of a particular factorization of Hilbert space.

Second, the probability problem. Deriving the Born rule probabilities within MWI is notoriously difficult. If all branches occur, in what sense are some outcomes more probable than others? Deutsch-Wallace decision-theoretic arguments and branch-counting approaches both face objections [Saunders et al., 2010].

Third, and most fundamentally: MWI denies that anything actually happens. No branch is more real than any other; the appearance of a definite outcome is an illusion relative to our branch. This conflicts with the most basic datum of experience—that something, specifically, occurs—and with the scientific methodology that requires comparing theory with a definite observational record. Relational Actualism preserves the ontological primacy of actual events: something happens, with probability given by the Born rule, and this happening dynamically eliminates specific possibilities. The wave function prunes rather than branches.

Fourth, the ontological parsimony argument cuts both ways. MWI is often praised for positing only one entity (the wave function) with no collapse. But it proliferates universes—an ontological cost not obviously lower than positing a single universe with genuine randomness.

5.2 Ontic Randomness, the Equal-Rate Limit, and the Kinematic Snap

The framework developed here implies that the determinism/indeterminism distinction is not fundamental but contextual. Nature is deterministic when the causal and logical context has reduced the possibilities for a variable to exactly one; it is irreducibly random when multiple possibilities remain viable.

It is crucial to note that while classical chaos in phase space represents purely epistemic uncertainty about a fully determined trajectory, the randomness in RAQM is *ontic*: the trajectory itself is physically undetermined until actualization dynamically prunes the space. This distinction between epistemic and ontic randomness is load-bearing for the framework and must not be elided.

The Kinematic Snap grounds ontic randomness precisely. The competing Poisson process derivation of the Born rule (Section 3.2.1) makes the locus of randomness explicit. Each volume element d^3x in the wavefunction support independently races to cross the kinematic threshold $p^\mu p_\mu = m^2c^2$. The time at which any given volume element crosses the threshold is governed by the quantum amplitude for the mediating boson to go on-shell — itself an underdetermined quantity until it occurs. This is not ignorance of a hidden variable. There is no fact of the matter, in the prior causal history, about when the threshold will be crossed at any specific location. The snap time at each location is ontic, not epistemic.

The equal-rate limiting case. A natural question arises: if two locations have exactly equal virtual exchange rates — as in a perfectly symmetric double-slit experiment or an equal-amplitude

two-state superposition — is there a fact of the matter about which location fires first? And if not, has determinism been smuggled back in through the back door?

The answer is no, and the argument is precise. In the equal-rate case, the competing Poisson process assigns $P(x_1) = P(x_2) = 1/2$. Neither location is favoured by the prior causal history. The snap at x_1 versus x_2 is not determined by any prior fact — it is a genuinely new, uncaused addition to the causal DAG. This is not a failure of the framework; it is the framework's central claim. The actualization event at x_0 is ontologically primitive: the transition from possible to actual is the irreducible locus of contingency in the universe.

To ask "but why x_0 rather than x_1?" is to ask for a prior cause of an event that, by definition, is the first link in a new causal chain. The question presupposes that every event must have a prior determining cause — which is precisely the deterministic assumption that ontic randomness rejects. In RA, the actualization event is not caused by prior events in the sense of being logically entailed by them. Prior events determine only the probability distribution over possible actualization locations. Which possibility becomes actual is the universe's irreducible freedom — the only place where the possible genuinely outstrips the determined.

Is perfect symmetry physically realizable? In practice, no experimental setup achieves perfect equality of virtual exchange rates at two locations. Environmental asymmetries, vacuum fluctuations, and the dense causal history of the laboratory break the symmetry at some level. This is not a loophole that restores determinism. Even with arbitrarily small asymmetry ε, the outcome at the less-favoured location occurs with probability $1/2 - \varepsilon/2$ — which approaches $1/2$ but never reaches zero. The randomness is not resolved by the asymmetry; it is merely shifted. The fundamental ontic randomness of the snap survives in every physically realizable case.

Comparison: why classical chaos is different. Classical chaos is deterministic at the level of individual trajectories. Given perfect initial conditions, the trajectory is fixed; only our ignorance of those conditions produces apparent randomness. In the RA Kinematic Snap, there are no hidden initial conditions that would determine the snap location if we knew them. The quantum amplitudes governing the threshold crossing time at each location are not waiting to be discovered — they are underdetermined until the snap occurs. The randomness is in the world, not in our description of it.

This is the distinction Bohr was groping toward but could not state precisely without the ontological framework RA provides. The wave function is not a description of our ignorance; it is a description of the universe's genuine openness. Actualization is the moment that openness closes.

Classical behavior is a limiting case, approached asymptotically as $\tau_D \to 0$ and $\langle \varepsilon_2 | \varepsilon_1 \rangle \to 0$. Large, warm, complex systems exhibit classical behavior not because they are fundamentally different from quantum systems, but because their dense network of actualization events has eliminated all degrees of freedom save one. This resolves the puzzle of the Heisenberg cut without invoking a cut at all.[1]

5.3 The Global Wave Function, Relativistic Causality, and Microcausality

There is one wave function for the universe [Hartle and Hawking, 1983]. Every interaction updates it. However, positing a global update to the wave function immediately invites a relativistic critique: a simultaneous global update seemingly requires a preferred universal 'now,' which would violate Lorentz invariance. This tension is resolved by formalizing the global wave function using the

[1] Schrödinger's cat deserves a brief defense—of the cat. The genuine locus of quantum indeterminacy in the thought experiment is the isotope: a small, isolated system whose decay is irreducibly random in exactly the sense RAQM identifies. A real cat is continuously actualizing—metabolizing, moving, emitting heat—a dense internal network of on-shell interactions that pins it firmly in the classical limit regardless of what the isotope is doing. RAQM rescues the world we actually see: the cat remains alive and indignant in its box until the decay event propagates through the mechanism and ends the matter. The paradox dissolves not because observation collapses a superposition, but because the cat was never in one.

Tomonaga-Schwinger equation. Rather than evolving according to a single global time parameter t, the universal quantum state is defined as a functional $\Psi[\sigma]$ on an arbitrary spacelike hypersurface σ:

$$i\hbar \frac{\delta \Psi[\sigma]}{\delta \sigma(x)} = H(x)\,\Psi[\sigma]. \tag{10}$$

An actualizing interaction updates the state functional along any valid spacelike hypersurface passing through x; the sequence of actualization events remains globally consistent regardless of how an observer slices spacetime.

The prohibition on superluminal signaling is established rigorously through algebraic microcausality. For any two spacelike-separated regions A and B, where $(x_A - x_B)^2 < 0$:

$$[\hat{O}_A, \hat{O}_B] = 0. \tag{11}$$

An actualization event at A instantly restricts the joint probability distribution—pruning possibilities within the global configuration space—but microcausality ensures that the *non-selective* (outcome-averaged) actualization map cannot causally force a measurable change in the expectation values of observables at spacelike-separated B. For a *conditioned* (post-selected) branch, the remote conditional state can in general differ from the pre-actualization state; however, because the specific outcome at A is not controllable from B, this conditional difference cannot be exploited to transmit information superluminally. There are genuine nonlocal constraints that are enforced whenever results are compared, a process that requires classical signaling at sub-luminal speeds.

A further complementary bound for non-relativistic systems: the Lieb-Robinson bound [Lieb and Robinson, 1972] establishes that in quantum lattice systems, the speed of propagation of correlations is bounded by an effective 'speed of light' determined by local interaction strength. This points toward a natural extension of the present framework to non-relativistic quantum systems—a direction pursued in the companion paper (Section 5.6), where the Causal Firewall defines the non-relativistic limit in which the local actualization history is universally agreed upon.

5.3.1 Consistency Condition for Actualization Events

The Tomonaga-Schwinger equation guarantees foliation-independence for unitary evolution: for any two spacelike-separated points x and y,

$$\left[\frac{\delta}{\delta\sigma(x)}, \frac{\delta}{\delta\sigma(y)}\right]\Psi[\sigma] = 0, \qquad (x-y)^2 < 0, \tag{12}$$

follows from microcausality. In free field theory this is the statement $[H(x), H(y)] = 0$ for $(x-y)^2 < 0$; in interacting, renormalized QFT the Hamiltonian density $H(x)$ must be understood as an operator-valued distribution whose local commutativity is one of the Wightman/Haag-Kastler axioms of algebraic QFT [Haag and Kastler, 1964], not an automatic consequence of naive density expressions. For the purposes of RAQM, we work within the framework of locally covariant QFT where local commutativity is assumed as an axiom; a fully constructive proof in specific interacting models (e.g., an Unruh-DeWitt detector coupled to a free scalar field) can be made explicit and would constitute a valuable concrete verification. What requires independent argument beyond this unitary foundation is whether RAQM's actualization map—the non-unitary operation that selects an outcome and updates $\Psi[\sigma]$ at an interaction vertex—also satisfies an appropriate consistency condition.

Let an actualization event occur at spacetime point x, and let σ_1 and σ_2 be any two spacelike hypersurfaces both passing through x, related by a local deformation in the causal future of x. The RAQM consistency condition requires that the probability distributions over all observable outcomes at any point y with $(x-y)^2 < 0$ be identical whether computed from $\Psi[\sigma_1]$ or $\Psi[\sigma_2]$ following actualization at x.

The argument proceeds in two steps. First, unitary evolution prior to actualization: the standard T-S consistency condition guarantees that $\Psi[\sigma_1]$ and $\Psi[\sigma_2]$ encode identical physical content up to

and including the actualization vertex at x. Second, the actualization map: RAQM's actualization is grounded in a local interaction at x satisfying the irreversible entanglement criterion, and therefore acts exclusively on the degrees of freedom in a neighbourhood of x. For any observable $\hat{O}_B$ supported in a region B spacelike-separated from x, algebraic microcausality gives $[\hat{O}_A, \hat{O}_B] = 0$, where $\hat{O}_A$ is any operator supported near x. It follows that the *non-selective* (outcome-averaged) actualization map—formally $\sum_k \mathscr{J}_k$ where $\{\mathscr{J}_k\}$ is the localized quantum instrument—leaves the expectation value of every observable at spacelike-separated B unchanged:

$$\mathrm{Tr}\left[\hat{O}_B \left(\sum_k \mathscr{J}_k\right)(\rho)\right] = \mathrm{Tr}[\hat{O}_B \rho]. \tag{13}$$

This is the rigorous no-signaling statement. It should be noted that for a *conditioned* (post-selected) outcome k, the remote conditional state $\rho_{B|k}$ can in principle differ from the pre-actualization ρ_B; no-signaling is preserved because the outcome k is not locally controllable from B. The foliation-invariance claim applies strictly to the non-selective map and to the resulting observable statistics, not to individual conditioned branches.

One further clarification. This argument establishes that observable statistics at spacelike-separated points are foliation-independent: no measurement at B can distinguish which hypersurface was used to carry the actualization at x. It does not establish that the global state functional $\Psi[\sigma]$ is literally identical on σ_1 and σ_2 following actualization—since actualization is a non-unitary operation, the global descriptions will in general differ. What is invariant is the empirical content: the full set of outcome probabilities accessible to any observer in the causal future of an actualization event. This is the physically relevant notion of foliation-independence for RAQM.

5.3.2 Foliation-Independence of the Kinematic Snap

The Kinematic Snap mechanism raises a sharpened version of the foliation-independence question. The competing Poisson process runs simultaneously across the entire wavefunction support: every volume element d^3x where $|\psi(\mathbf{x})|^2 > 0$ is independently racing to cross the kinematic threshold. Different spacelike hypersurfaces will slice this distributed, simultaneous process differently. We must verify that the snap location and the Born probabilities are hypersurface-independent.

The snap event is a spacetime point, not a hypersurface event. The first and most important observation is that the Kinematic Snap, when it occurs, is a local event at a specific spacetime point x_0^μ. A virtual exchange at x_0^μ crosses the threshold $p^\mu p_\mu = m^2c^2$ and an on-shell boson is emitted from x_0^μ. This is a pointlike event in spacetime — it has a definite location in all reference frames, because spacetime points are covariant. Different observers may assign different spatial coordinates and different times to x_0^μ, but they agree that the snap occurred there.

The snap is therefore not "which location fires first on a given hypersurface" but rather "which spacetime point is the first to emit an on-shell boson." The ordering is causal, not hypersurface-dependent.

The competing Poisson process in covariant form. The snap rate at spacetime point x^μ is proportional to the local probability density in the rest frame of the interaction. For a Dirac field with conserved four-current $j^\mu = \bar{\psi}\gamma^\mu\psi$, the timelike component $j^0 = \psi^\dagger\psi = |\psi(x)|^2$ gives the probability density in the local rest frame. The Lorentz-covariant snap rate is

$$r(x^\mu) \;\propto\; j^\mu(x)\, u_\mu(x), \tag{14}$$

where $u^\mu(x)$ is the future-directed unit timelike vector of the local rest frame. In the rest frame, $u^\mu = (1, \mathbf{0})$ and $r \propto j^0 = |\psi(x)|^2$. Note that the contracted scalar $j^\mu j_\mu$ is a Lorentz invariant but does not equal $|\psi|^2$ in general; the physically relevant quantity is the rest-frame probability density $j^\mu u_\mu$.

The competing process is therefore between spacetime volume elements d^4x, not between spatial volume elements d^3x on a preferred hypersurface. The probability that the snap occurs in d^4x at x^μ before it occurs anywhere else in the support of the wavefunction is:

$$dP(\text{snap at } x^\mu) = \frac{r(x^\mu)\, d^4x}{\int_{\text{supp}(\psi)} r(x'^\mu)\, d^4x'}, \tag{15}$$

where the denominator integrates over the spacetime support of ψ, ensuring the normalisation is finite and well-defined. The snap location is determined by the four-dimensional local probability density, not by a preferred time coordinate.

Born rule is foliation-independent. For a non-relativistic wavepacket localized in time (a pulse of finite duration), the integration over dt in the denominator is the same for all observers up to time-dilation factors that cancel in the ratio. For each spatial location, the probability of snap at $\mathbf{x}$ integrates to $|\psi(\mathbf{x})|^2$ regardless of which spacelike hypersurface is used to evaluate it, because the time-integrated rate at each spatial location is proportional to $|\psi(\mathbf{x})|^2$ in every frame. The Born rule is therefore foliation-independent.

The snap time is frame-dependent; the snap event is not. Different observers will assign different times t_0 to the snap event at x_0^μ. An observer moving relative to the apparatus will see the snap as occurring earlier or later. But they will agree on: (a) that the snap occurred at x_0^μ; (b) that the Born probability for x_0^μ is $|\psi(x_0)|^2$; and (c) that the wavefunction evolves from x_0^μ onward. The snap time is frame-dependent in the same sense that any spacetime coordinate is frame-dependent. The physical content of the snap — the vertex written to the DAG, the on-shell emission, the irreversible entanglement record — is a covariant fact.

What remains open. The argument above establishes foliation-independence for the snap location and Born probabilities in perturbative QFT where the competing Poisson process approximation is valid. Two refinements require further work:

1. **Entangled multi-particle snaps.** When two entangled particles interact with separate detectors at spacelike separation, the snap at detector 1 and the snap at detector 2 must be shown to be consistent regardless of which snap is described as occurring "first" in any given frame. The no-signaling result (Section 5.3.1) guarantees that observable statistics are frame-independent, but the joint snap probability for entangled pairs requires a careful treatment using the covariant four-current density for the joint system rather than individual single-particle currents.

2. **Non-perturbative regime.** The Poisson process approximation assumes the snap events are rare and independent. In strongly coupled regimes, the snap events may not be statistically independent, and the covariant rate formula requires the full AQFT formulation. The finite-dimensional version of the required frame-independence result has now been formally verified in Lean 4: the theorem `frame_independence` in `RA_AQFT_Proofs_v10.lean` establishes that $S(U\rho U^\dagger \| \sigma_0) = S(\rho \| \sigma_0)$ for any unitary U, using the unitary-invariance of relative entropy together with Poincaré-invariance of the vacuum as a physics axiom. The proof uses the `Matrix.cfc_conj_unitary` lemma from the Lean-QuantumInfo library [Meiburg et al., 2025]. The remaining target is the infinite-dimensional extension: showing that $\mathscr{P}_{\text{act}}$, defined via local algebras, commutes with wedge algebra modular flow in the Tomita-Takesaki sense.

The second open item (non-perturbative regime / full AQFT extension) has been partially resolved: the frame-independence claim is now a proved theorem in finite dimensions. The physical picture is established and the finite-dimensional Lean verification is complete; the remaining target is the infinite-dimensional algebraic QFT extension to type III_1 von Neumann algebras. A collaborator with expertise in Tomita-Takesaki modular theory would be well-positioned to complete this step.

5.4 Free Will, Creativity, and the Constrained Observer

A provocative implication of the framework deserves careful treatment. RAQM forecloses libertarian free will as causally incoherent. An uncaused cause would be a causal contradiction in a universe whose consistency is guaranteed by the relational structure of actualization. Every actualizable property is intrinsically relational: it has no determinate value except in the context of a specific causal interaction. A choice that was causally independent of all prior context would be a property with a determinate value independent of all relational context—which is exactly what RAQM rules out as ontologically incoherent.

What RAQM does preserve is genuine creativity. The universe described by RAQM is not a clockwork in which all outcomes are fixed in advance. At the quantum level, actualization outcomes are irreducibly random when multiple possibilities remain viable. This randomness is not mere noise—it is the universe's genuine openness to novelty. Each new actualization event adds to the causal record of the universe in ways that were not fixed by prior history. Freedom, understood as the genuine openness of the future and the real causal efficacy of minds in shaping which possibilities are actualized, is not merely compatible with RAQM—it is demanded by it.

5.5 Empirical Support and Falsifiability

A foundational critique often leveled at ontological interpretations of quantum mechanics is the problem of empirical equivalence. Relational Actualism goes beyond empirical equivalence by positing that actualization is a continuous, physical process of possibility-elimination occurring over a finite duration, driven by causal interactions, rather than an instantaneous mathematical discontinuity.

This prediction is actively supported by recent experimental breakthroughs. Measurements of artificial atoms in superconducting circuits have demonstrated that quantum jumps are not instantaneous, discontinuous events, but continuous and even reversible transitions occurring over finite timescales [Minev et al., 2019]. Similarly, attosecond chronoscopy of photoemission [Pazourek et al., 2015] has shown that the duration of an electron's transition is strictly governed by the local structural and interaction environment. These results align with RAQM's mechanism, where actualization is governed by the continuous decay of off-diagonal coherences ($\langle \varepsilon_2 | \varepsilon_1 \rangle \to 0$) and bounded by local interaction strengths.

By anchoring the transition of the possible to the actual in these measurable, finite dynamics, Relational Actualism distinguishes itself empirically from instantaneous collapse models. We now state the specific predictions more precisely.

Prediction 1: Finite actualization propagation velocity. RAQM predicts that the actualization front — the boundary between the superposed and actualized regions of a many-body system — propagates at a finite velocity bounded by the local interaction strength, not instantaneously. In a quantum lattice system with nearest-neighbor coupling J, the Lieb-Robinson bound [Lieb and Robinson, 1972] establishes that correlations propagate at most at velocity $v_{\mathrm{LR}} \sim J \cdot \ell / \hbar$, where ℓ is the lattice spacing. RAQM predicts that the actualization front propagates at or below this bound. The testable deviation from instantaneous collapse models: in a spatially extended many-body system, the decoherence of a distant site B following an actualization event at site A should exhibit a time delay $\Delta t \geq |r_A - r_B| / v_{\mathrm{LR}}$ before the off-diagonal coherences at B begin to collapse. Instantaneous collapse models predict no such delay.

Prediction 2: Environment-dependence of transition duration. RAQM predicts that the duration of a quantum transition is not an intrinsic property of the system alone, but is governed by the local environmental coupling strength. Specifically, the actualization timescale τ_{act} should scale as $\tau_{\mathrm{act}} \sim \hbar / \Delta E_{\mathrm{env}}$, where ΔE_{env} is the characteristic energy scale of the environmental interaction driving the entanglement. This is consistent with the Pazourek et al. attosecond chronoscopy

results [Pazourek et al., 2015], where the ionization duration depends on the local Coulomb environment, and with Minev et al.'s demonstration that the jump duration depends on the coupling to the measurement apparatus. The prediction is that systematically varying the environmental coupling in a controlled superconducting circuit experiment should produce a proportional variation in transition duration — a clean, quantitative test.

Prediction 3: Reversibility as a diagnostic of actualization threshold. RAQM distinguishes pre-actualization dynamics (reversible, because the thermodynamic horizon has not been crossed) from post-actualization dynamics (irreversible, because a permanent causal record has been established). The quantum eraser experiments [Scully and Drühl, 1982] and the Minev et al. jump-reversal results [Minev et al., 2019] both demonstrate that what appears to be a collapse can be reversed if the environmental record is erased before thermodynamic irreversibility is established. RAQM predicts a sharp threshold between these two regimes, determined by the environmental entanglement entropy crossing a critical value. Below threshold: the transition is reversible in principle. Above threshold: the causal record is permanent and the transition cannot be reversed regardless of intervention. Identifying this threshold experimentally — the precise point at which reversal becomes impossible — would constitute a direct measurement of the actualization criterion.

Distinguishing RAQM from competing interpretations. All three predictions above are shared by RAQM and *not* by the following alternatives: (a) instantaneous collapse models (GRW, CSL) predict no propagation delay and no environment-dependence of the collapse threshold beyond the stochastic rate parameter; (b) Everettian Many-Worlds predicts no sharp threshold and no genuine irreversibility — only apparent irreversibility from within a branch; (c) decoherence-only models predict environment-dependence of decoherence timescales but deny that a sharp actualization threshold exists. RAQM's predictions are therefore genuinely discriminating, not merely reframings of existing results.

5.6 Future Directions: Curved Spacetime and the Arrow of Time

While the present framework resolves the measurement problem within the domain of flat Minkowski spacetime, grounding actualization in irreversible environmental entanglement naturally invites extension into non-inertial reference frames and curved spacetime. Because the definition of an on-shell excitation is frame-dependent via the Unruh and Hawking effects, RAQM implies that the rate and physical reality of wave function pruning must be inherently frame-dependent, deeply tying the structure of actualization to the causal horizons of the observer.

Furthermore, by defining actualization as the irreversible elimination of possibilities, this framework provides a purely ontological grounding for the arrow of time. The formal extension of Relational Actualism to causal severance at event horizons and non-inertial thermodynamics is developed in the companion paper *Relational Actualism Gravity and Cosmology* (RAGC) [Sandeman, 2026b]. That paper proves two results that flow directly from the actualization criterion established in Section 3.2.1 of the present work: first, that the RA-modified stress-energy operator has zero vacuum expectation value, structurally eliminating the vacuum energy contribution to the metric source term; and second, that for any causal severance of the spacetime DAG, the local conservation condition holds independently on each disconnected subgraph, formally establishing the energy bookkeeping dissolution across event horizons. The second result—the Ledger Partition Theorem—has been **formally verified in Lean 4/Mathlib**, establishing the logical soundness of its discrete graph-theoretic core.

RAGC further makes three empirical predictions that follow structurally from the actualization criterion established here: that the inferred value of H_0 correlates with the mean actualization density along the line of sight (providing a geometric mechanism for the Hubble Tension without dark energy); that the vacuum energy contribution to the metric is structurally zero rather than fine-

tuned; and that daughter spacetimes nucleated at causal severances inherit an effective cosmological constant determined by the local actualization flux at nucleation.

The companion paper *Relational Actualism Complexity and Intelligence* (RACI) [Sandeman, 2026c] extends the framework to biological complexity and agency, proving that the Markov blanket isomorphism and the thermodynamic steering cost bound are structurally grounded in the same actualization criterion, and making four empirical predictions including a physically derived constraint on the distribution of life in the universe. Core theorems in RACI have been formally verified in Lean 4/Mathlib. The load-bearing Condition C1′ of the Assembly Index Correspondence theorem—that each JOIN operation in molecular assembly corresponds to at least one irreversible actualization event (a coarse-grained lower bound robust to condensed-phase polymer chemistry)—has been established by a variational argument (the variational principle guarantees all stable covalent bond formation is exothermic, mandating irreversible energy dissipation to satisfy the Local Ledger Condition) and confirmed by B3LYP/6-311+G* density functional calculations across the five primary bond types of biological assembly (C-C, C-N, C-O, S-S, P-O), all verified exothermic. The Assembly Index Correspondence theorem therefore holds across the full chemical alphabet of biological complexity.

6 Conclusion

Einstein put himself on the path to significant discoveries by accepting that the speed of light is a constant of Nature and working out the consequences. What he could not accept was a universe where true randomness is possible. Yet this is what nature really seems to be like. Relational Actualism accepts indeterminacy and works out its consequences.

Properties are of three kinds: invariant (always actualized), classical (actualized deterministically through interaction), and intrinsically relational (actualized only in specific contexts of interaction, with genuine ontic randomness when more than one possibility remains). Crucially, accepting this intrinsic randomness does not require abandoning relativistic causality.

By formalizing the global wave function using the Tomonaga-Schwinger equation within the Hartle-Hawking framework, the wave function of the universe is real, dwelling in a Hilbert space whose geometry (as evidenced by the Berry phase) is physically meaningful. Possibilities and actualities are both real; manifest reality is the constant process of the possible becoming actual through interaction—a dynamic breathing of the local phase space volume, where degrees of freedom are physically pruned and, prior to thermodynamic irreversibility, can be restored.

Because this global actualization is foliation-independent—as established by a derived consistency condition showing that RAQM's actualization map leaves observable statistics at spacelike-separated points invariant under hypersurface choice (equation 13), applying rigorously to the non-selective map—and rigorously bounded by microcausality, it enforces nonlocal logical constraints in configuration space without ever permitting superluminal signaling in Minkowski spacetime.

This framework distinguishes itself from Relational Quantum Mechanics by grounding actualization in causal interaction rather than perspectival observation. It distinguishes itself from dynamical collapse models by grounding the collapse trigger in irreversible environmental entanglement rather than a stochastic threshold or new fundamental constant. It distinguishes itself from decoherence-only approaches by adding the ontological claim that possibility-elimination is genuine, not merely apparent. And it distinguishes itself from Many-Worlds by preserving the primacy of actual events and the reality of a single history.

The universe may not always know what it is going to do, but it always knows what it has done—and this non-locally constrains what is then possible, progressively funneling its unconstrained degrees of freedom into a definitive, classical record. There is a simple, if profound, logic to all of it. One would think that Einstein would have liked that.

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Code availability: The Lean 4 formal proof files (`RA_AQFT_Proofs_v10.lean` and `RA_Proofs_Lean4.lean`), Python computational scripts, and all companion paper source files are publicly available at github.com/jsandeman/Relational-Actualism.

Data availability: This paper presents theoretical work. No datasets were generated or analysed. All formal verification results are contained in the Lean 4 source files available at the repository above.

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